

Heuristic Short-term Path Prediction for Spontaneous Human Locomotion in Virtual Open Spaces

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ABSTRACT

Redirected Walking (RDW) shrinks large virtual environments to fit small physical tracking spaces while supporting natural locomotion. Particularly in predictive RDW, one of the core concepts of RDW, algorithms rely on predicting users' future paths to adjust the induced redirection, which manipulates users' perception to deviate their physical paths from the intended virtual paths. Current path predictions either assume drastic simplifications or build on complex human locomotion models, which are inappropriate for real-time planning and thus not usable for RDW. Further, adapting existing predictive RDW algorithms to unconstrained open space exponentially increases their computational complexity, so that they are not applicable in real-time.

In this work-in-progress paper, we discuss the currently prevalent issues of path prediction in RDW and propose simple yet flexible path prediction models that support dynamic virtual open spaces. Our proposed prediction models consist of two shapes: a drop shape represented by the lemniscate of Bernoulli and a sector shape. They define an area, in which linear and clothoid walking trajectories will be investigated.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Redirected Walking (RDW) allows a user to explore extensive virtual environments by real walking in a constrained physical tracking space [8]. Where real-walking implies that motion in reality is mirrored in the virtual environment, RDW situationally modifies this direct mapping to actively deviate the user from a real-world path, while still perceiving the intended virtual path. Depending on the redirection application, RDW can be divided into three core concepts: scripted (i.e., fixed path), reactive (i.e., no prior path knowledge), and predictive (i.e., real-time path prediction). Predictive approaches for RDW have been least investigated, supposedly due to their complexity and error proneness caused by human spontaneity. Such approaches depend on reliable and accurate estimation of future paths to determine required and valid redirection over time and most of them operate on simplifications such as limiting the potential virtual paths a user can follow or requiring stationary virtual environments. The resulting virtual environments thus often consist

of sequences of static corridors or mazes with little room for individual deviation or exploration [7], which consequently also excludes their use for multiple co-located users. Because of exponentially growing computational complexity of redirection planning, previous efforts have not transferred predictive approaches to unconstrained, virtual open spaces.

In this paper, we suggest a concept for short-term path prediction which uses simple prediction models to support real-time predictive RDW applications in virtual open spaces. For the remainder of the paper, we first provide an overview of previous approaches to path prediction and planning walking trajectories. Afterwards, we derive our new locomotion models that predict the immediate next steps of a user in a virtual environment due to their short-termed nature.

2 RELATED WORK

Prediction in Redirected Walking

The user's walking trajectory can be altered by applying redirection techniques such as linear, rotational or curvature gains in the virtual environment [8], which effectively compress extensive virtual environments into smaller physical spaces. The core question in RDW is which technique to apply and with which intensity, as this depends on the virtual scenario. Generally, there are three concepts for choosing redirection techniques: reactive, predictive, and scripted.

Since the focus of this paper is on prediction, we only discuss related examples of predictive RDW. A popular example thereof is the algorithm FORCE, which creates a skeleton graph consisting of nodes and segments that describe the walkable paths in virtual corridors [12]. Following the segments of the graph, a predicted path is generated for the redirection calculation. Alternatively, MPCRed recursively evaluates a cost function consisting of actions such as applying redirection, while also following a similarly predefined bi-directional skeleton graph [7].

Human Locomotion in Virtual Reality

Arechavaleta et al. [1] introduced an early approach towards estimating human walking behavior, observing human navigation and comparing it to basic robotic locomotion. They derived a differential system that describes human locomotion in a real, open space environment, characterized by a non-holonomic walking behavior that relies on the body orientation of a person to always be tangential to the walking direction. Hicheur et al. [5] investigated the similarity of velocity profiles and the path geometries of a goal-oriented task and reported stereotypical properties for human locomotion in their controller user study. Another locomotion model introduced by Cirio et al. [3] uses Euler integration for positional updates based on the observation of the linear relation between position and walking velocity. Finally, Zank & Kunz [10] introduced a locomotion model relying on multiple predefined locomotion targets, creating a set of predefined trajectories that are continuously compared against the actual user trajectory. They later showed how skeleton graphs can be effectively extracted from virtual environments and further

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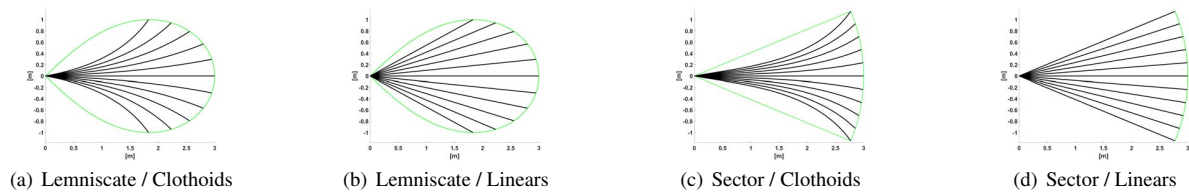


Figure 1: The four proposed models to capture the immediate next steps of users exploring a virtual environment. The exact scaling in horizon is subject of current investigation will most likely be related to the current walking speed of the users.

created locomotion targets in a continuous manner [11]. Overall, many locomotion models have been introduced so far, but they either lack real-time capabilities due to computational complexity or the approaches depend on virtual constraints (e.g., static mazes or corridors) to create more or less reliable skeleton graphs.

3 MODEL DERIVATION

Predictive RDW approaches rely on long-term prediction to maximize the effect of the planning procedure [7]. Even though Zank showed that long-term predictions are possible to a certain degree, transferring such predictions to large open spaces renders them unstable and error-prone [9]. Further, with increasing prediction horizon, the computational complexity of the prediction planner grows exponentially. To mitigate the effect of faulty predictions and account for human spontaneity, we suggest reducing the prediction horizon to a short-term approach in favor of increasing the frequency of calculations of redirection. By limiting the horizon to the short-term, prediction is reduced to an immediate specific region of interest around the user. According to Guitton et al. [4], users primarily prioritize following their field of view while navigating. The effective field can be further narrowed, since users' saccadic gaze shifts are mostly limited to 45° to the left or right [4] from the user's current view center. Having established this region of interest, we focus on the immediate area in front of a user and seek potential descriptions for the shape of the area and the trajectories therein. First, we develop a base shape that our prediction will assume within the region of interest. Building on a conceptual and unvalidated suggestion for short-term path prediction [6], we formalize a drop-like shape using the mathematical description of the lemniscate of Bernoulli (see Fig. 1(a) and Fig. 1(b)). Further, a common tool used in RDW is the Redirected Walking Toolkit [2], which includes a synthetic path generation. The toolkit relies on the assumption that motion is executed sequentially. Accordingly, simulated users always turn on spot, move straight and then turn on spot again. Applying this concept to our prediction and pairing it with our region of interest, we form a 90° sector of a circle with the origin coinciding with the user (see Fig. 1(c) and Fig. 1(d)). Second, we discuss the type of the trajectories within the region of interest. Following conceptual suggestions [6] and the toolkit's synthetic paths [2], we initially assume linear trajectories. Because human locomotion generally shows a non-holonomic nature [1], it is however better approximated by geodesics from assembled clothoidic arcs. Clothoidic trajectories change their curvature in relation to the length of the trajectory and are defined by their origin, their initial curvature and their length. In our case, the origin is given by the user's position, we limit the initial curvature to a bandwidth of $\pm 10^\circ$ for simplicity, and truncate the length at the intersection with the respective prediction shape.

4 CONCLUSION & OUTLOOK

In this work-in-progress paper, we address a prevalent issue limiting predictive RDW. In recent approaches, predictions were simplified by virtually constrained, static environments which allow extracting skeleton graphs prior to user exploration. However, in case users

step into virtual open spaces or engage in dynamic environments, predictors fail or lead to an overload of the planning algorithms. Accordingly, an effective and simple short-term path prediction which can be deployed both in virtually constrained and open spaces will greatly benefit the development of new predictive RDW approaches. We aim to pursue the presented concepts and validate them in an experimental user study to identify their performance. For this, adequate user data shall be acquired from virtual environments covering the spectrum from virtually constrained to completely open spaces. So far, the scaling of the locomotion models regarding the horizon is still investigated and will be subject of the experimental evaluation. A promising approach may relate the horizon depth with the walking speed to adjust for variations thereof.

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